



Amazon Route 53

Lecture Slides

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Good luck on your exam!

Route 53 – Introduction



Route 53 is a **Domain Name Service (DNS)**

think GoDaddy or NameCheap but with integrations with AWS Services.



Selected domains (1/5)

Domain registration fee

bathbombsformoms.io

71.00 USD

Remove

Subtotal:

71.00 USD

The domain registration fee displayed is for 1 year. You can change duration on the next page.

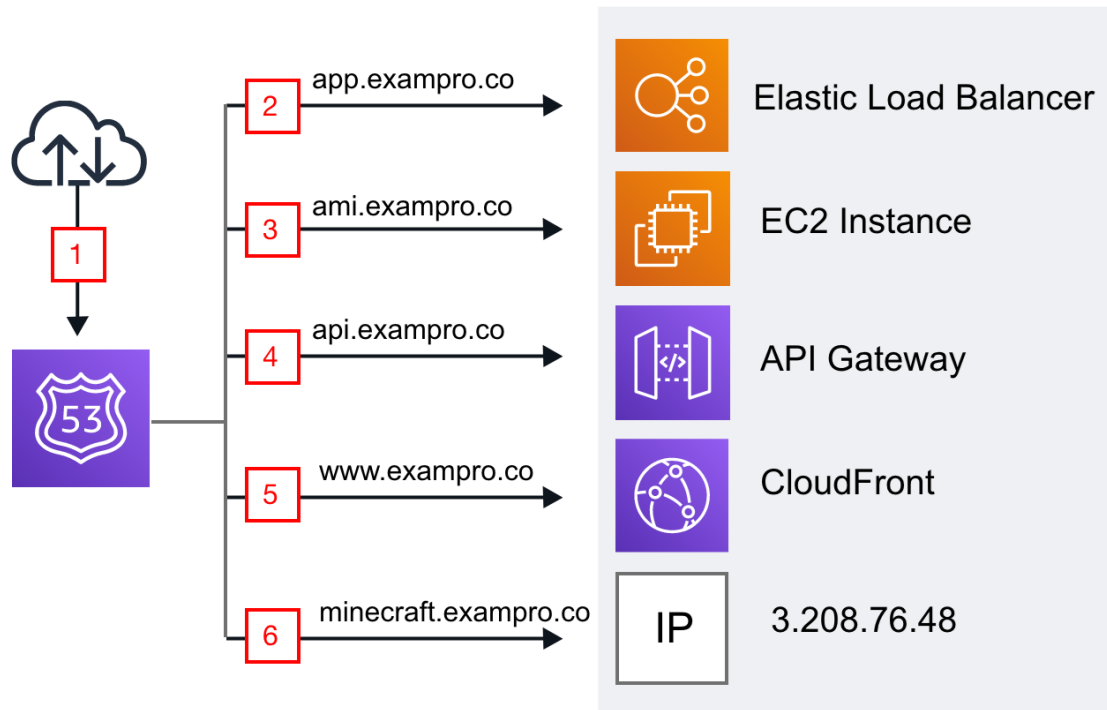
Proceed to checkout

You can:

- **Register** and manage domains
- Create various record sets on a domain
- Implement complex traffic flows eg. Blue/green deploy, failovers
- Continuously monitor records via health checks
- Resolve VPC's outside of AWS

Route 53 - Use Case

Use Route 53 to get your custom domains to point to your AWS Resources



1. Incoming internet traffic
2. Route traffic to our web app backed by ALB
3. Route traffic to an instance we use to tweak our AMI
4. Route traffic to API gateway, which powers our API
5. Route traffic to CloudFront which serves our S3 static-hosted website
6. Route traffic to an Elastic IP (EIP) which is a static IP that hosts our company's Minecraft server

Route 53 – Hosted Zone

A **hosted zone** is a **container for record sets**, scoped to route traffic for a specific domain or subdomains

There are two types of Zones:

- **Public** Hosted Zones — How you want to route traffic inbound from the **Internet**
- **Private** hosted Zones — How you want to route traffic within **an Amazon VPC**



```
aws route53 create-hosted-zone \  
--name example.com \  
--caller-reference "2014-04-01-18:47" \  
--hosted-zone-config Comment="Main Domain"
```



```
aws route53 create-hosted-zone \  
--name vpc.example.com \  
--vpc VPCRegion=ca-central-1,VPCId=vpc-c3be22b9 \  
--caller-reference "2014-04-01-18:47" \  
--hosted-zone-config PrivateZone=true
```

Route 53 – Hosted Zone

Let's say you wanted to build a SaaS product where each customer had their own subdomain on the app subdomain.

eg. **mycustomer1**.app.mysaasapp.com

Create a hosted zone domain

- Create a record set to the subdomain pointing to the nameservers of the subdomain

Create a hosted zone for the subdomain eg. App

- Create a record set for the wildcard to send traffic back to the root of the subdomain

```
Resources:
  DomainHZ:
    Type: AWS::Route53::HostedZone
    Properties:
      Name: example.com
  SubdomainHZ:
    Type: AWS::Route53::HostedZone
    Properties:
      Name: app.example.com
  RecordSet:
    Type: AWS::Route53::RecordSet
    Properties:
      HostedZoneId: !Ref DomainHZ
      Name: app.example.com
      Type: NS
      TTL: '300'
      ResourceRecords: !GetAtt SubdomainHZ.NameServers
  WildcardRecordSet:
    Type: AWS::Route53::RecordSet
    Properties:
      HostedZoneId: !Ref SubdomainHZ
      Name: "*.app.example.com"
      Type: A
      ResourceRecords:
        - "app.example.com."
```

Route 53 – Record Sets

Record Sets are a collection of records which determine where to send traffic

Record Sets are always **changed in batch** via the API

```
aws route53 change-resource-record-sets \
--hosted-zone-id YOUR_HOSTED_ZONE_ID \
--change-batch '{"Changes": [{
  "Action": "UPSERT",
  "ResourceRecordSet": {
    "Name": "blog.example.com",
    "Type": "CNAME",
    "TTL": 300,
    "ResourceRecords": [{
      "Value": "news.example.com"
    }]
  }
}]}'
```



The following record types supported by Route53

- A record type
- AAAA record type
- CAA record type
- CNAME record type
- DS record type
- MX record type
- NAPTR record type
- NS record type
- PTR record type
- SOA record type
- SPF record type
- SRV record type
- TXT record type

There are three action types:

- **CREATE** : Creates a resource record set that has the specified values.
- **DELETE** : Deletes an existing resource record set that has the specified values.
- **UPSERT** : If a resource set doesn't exist, Route 53 creates it. If a resource set exists Route 53 updates it with the values in the request.

Route 53 – Alias Record

AWS has its own special **Alias Record** which extends DNS functionality. It will route traffic to specific AWS resources.

Alias records are smart where they can detect the change of an IP address and continuously keep that endpoint pointed to the correct resource.

In most cases, you want to be using **Alias** when routing traffic to AWS resources

Alias Records are Type A records with the **Alias Target** configured

```
aws route53 change-resource-record-sets \
--hosted-zone-id "Z3AQBSTGFYJSTF" \
--change-batch '{"Changes": [{
  "Action": "UPSERT",
  "ResourceRecordSet": {
    "Name": "example.com",
    "Type": "A",
    "AliasTarget": {
      "DNSName": "s3-website-us-east-1.amazonaws.com",
      "HostedZoneId": "Z3AQBSTGFYJSTF"
    }
  }
}]}'
```



Alias Target can point to:

- CloudFront — d111111abcdef8.cloudfront.net
- Elastic Beanstalk environment — example.elasticbeanstalk.com
- ELB load balancer — example-1.us-east-2.elb.amazonaws.com
- S3 website endpoint — s3-website.us-east-2.amazonaws.com
- Resource record set — www.example.com
- VPC endpoint — example.us-east-2.vpc2.amazonaws.com
- API Gateway Endpoint custom regional API — d-abcde1234.execute-api.us-west-2.amazonaws.com

Route 53 – Routing Policies

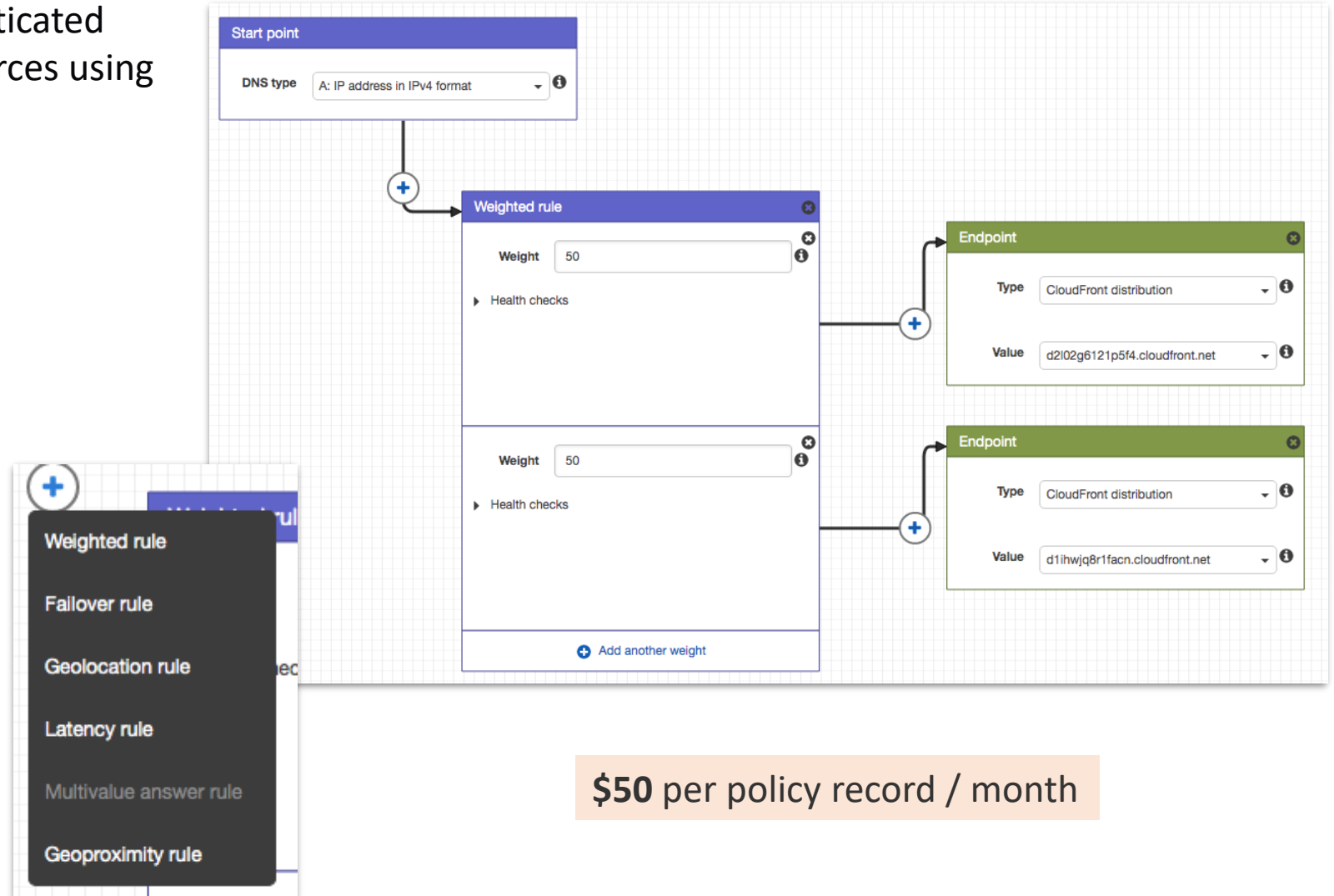
There are **8 different types** of Routing Policies available inside Route53

- **Simple Routing:** Default routing policy; if multiple IP addresses are available, selection is **random**
- **Weighted Routing:** Distributes traffic based on assigned **weights** (e.g., 70% to one resource, 30% to another)
- **Latency-Based Routing:** Directs traffic to the **region** with the lowest **latency** for the user
- **Failover Routing:** Routes traffic to a **secondary endpoint** if the **primary** is **unhealthy**
- **Geolocation Routing:** Routes traffic based on the **user's location**
- **Geoproximity Routing** Routes traffic based on the **resource location** and can **shift traffic** from one region to another
- **IP-Based Routing:** Routes traffic based on the **user's IP address**. Allows granular control by uploading **user-IP-to-endpoint mappings** to Route 53. Useful for optimizing performance or reducing network costs by directing users from specific **IP ranges** or **ISPs** to designated endpoints
- **Multi-Value Answer Routing:** Returns up to **8 healthy records** in response to a **DNS query**, selected at random

Route 53 – Traffic Flow

A **visual editor** lets you create sophisticated routing configurations for your resources using existing routing types

Supports **versioning** so you can roll out or roll back updates.



\$50 per policy record / month

Route 53 – Simple Routing Policies

Simple Routing Policies are the most basic routing policies in Route53 **Default Policy**

- You have 1 record and provide multiple IP addresses
- When multiple values are specified for a record, Route53 will return all values back to the user in a **random order**

For example, if you had a record for 'www.exampro.co' with 3 different IP address values, users would be directly **randomly to 1 of them** when visiting the domain.



```
aws route53 change-resource-record-sets \
--hosted-zone-id YOUR_HOSTED_ZONE_ID \
--change-batch '{"Changes": [{
  "Action": "UPSERT",
  "ResourceRecordSet": {
    "Name": "www.exampro.co",
    "Type": "A",
    "TTL": 300,
    "ResourceRecords": [
      { "Value": "34.229.79.211" },
      { "Value": "18.212.245.88" },
      { "Value": "3.208.76.48" }
    ]
  }
}]}'
```

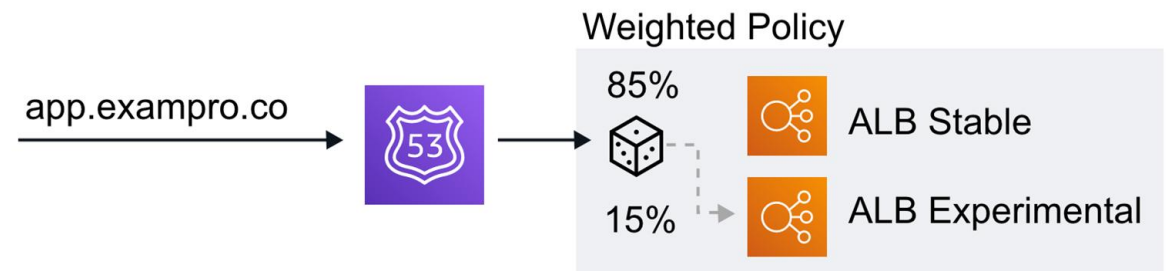
Route 53 - Weighted Routing Policies

Weighted Routing Policies let you split up traffic based on different 'weights' assigned.

This allows you to send a certain percentage of overall traffic to one server and have any other traffic to be directed to a completely different server.

```
{
  "Changes": [{
    "Action": "UPSERT",
    "ResourceRecordSet": {
      "Name": "app.exampro.co",
      "Type": "A",
      "TTL": 300,
      "WeightedRoutingPolicy": {
        "WeightedRecords": [
          { "Value": "34.229.79.211", "Weight": 85 },
          { "Value": "18.212.245.88", "Weight": 15 },
        ],
        "FallbackBehavior": "NON_FAILOVER"
      }
    }
  ]
}
```

For example, if you had an ALB running experimental features, you could test against a small amount of traffic at random to minimize the impact of effect

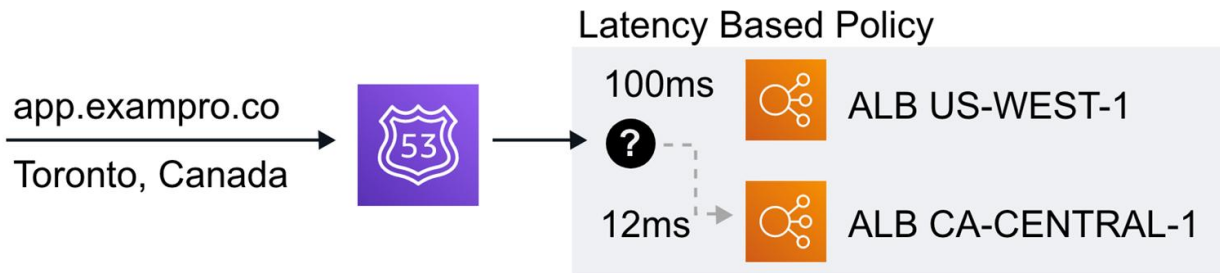


Route 53 – Latency Based Routing Policies

Latency Based Routing allows you to direct traffic based on the lowest network latency possible for your end-user **based on region**.

Requires a latency resource record to be set for the EC2 or ELB resource that hosts your application in each region.

For example, you have two copies of your web app backed by ALB. One in California, US and another in Montreal, Canada. A request comes in from Toronto will be routed to Montreal since it will have lower latency



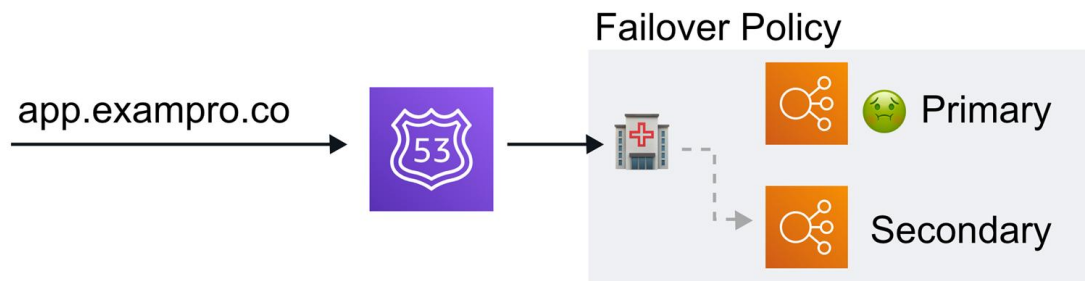
```
{
  "Changes": [{
    "Action": "UPSERT",
    "ResourceRecordSet": {
      "Name": "example.com",
      "Type": "A",
      "AliasTarget": {
        "DNSName": "dualstack.myalb1.us-east-1.elb.amazonaws.com",
        "EvaluateTargetHealth": false,
        "HostedZoneId": "YOUR_LOAD_BALANCER_1_HOSTED_ZONE_ID"
      },
      "SetIdentifier": "us-east-1",
      "Region": "us-east-1",
      "LatencyRoutingPolicy": { "Region": "us-east-1" }
    },
    "SetIdentifier": "us-east-1",
    "Region": "us-east-1",
    "LatencyRoutingPolicy": { "Region": "us-east-1" }
  }],
  {
    "Action": "UPSERT",
    "ResourceRecordSet": {
      "Name": "example.com",
      "Type": "A",
      "AliasTarget": {
        "DNSName": "dualstack.myalb2.us-west-1.elb.amazonaws.com",
        "EvaluateTargetHealth": false,
        "HostedZoneId": "YOUR_LOAD_BALANCER_2_HOSTED_ZONE_ID"
      },
      "SetIdentifier": "us-west-1",
      "Region": "us-west-1",
      "LatencyRoutingPolicy": { "Region": "us-west-1" }
    },
    "SetIdentifier": "us-west-1",
    "Region": "us-west-1",
    "LatencyRoutingPolicy": { "Region": "us-west-1" }
  },
  // ...
}
```

Route 53 – Failover Routing Policies

Failover Routing Policies allow you to create active/passive setups in situations where you want a primary site in one location and a secondary data recovery site in another.

Route 53 automatically monitors health checks from your primary site to determine the health of end-points. If an end-point is determined to be in a failed state, all traffic is automatically directed to the secondary location.

For example, we have a primary and secondary web app backed by ALB. Route 53 determines our primary is unhealthy and fails over to secondary ALB.

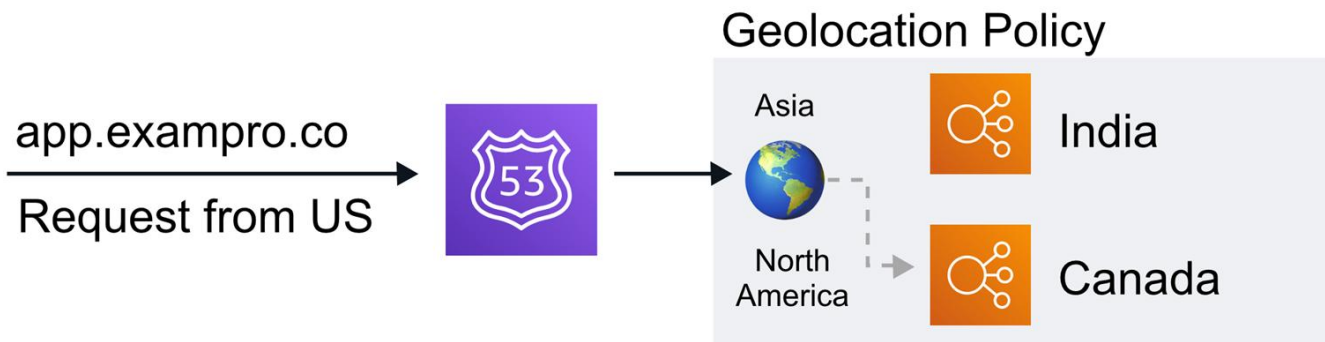


```
{
  "Changes": [
    {
      "Action": "UPSERT",
      "ResourceRecordSet": {
        "Name": "example.com",
        "Type": "A",
        "SetIdentifier": "Primary",
        "FailoverRoutingPolicy": {
          "FailoverBehavior": "PRIMARY"
        },
        "TTL": 300,
        "ResourceRecords": [
          {
            "Value": "34.229.79.211"
          }
        ]
      }
    },
    {
      "Action": "UPSERT",
      "ResourceRecordSet": {
        "Name": "example.com",
        "Type": "A",
        "SetIdentifier": "Secondary",
        "FailoverRoutingPolicy": {
          "FailoverBehavior": "SECONDARY"
        },
        "TTL": 300,
        "ResourceRecords": [
          {
            "Value": "18.212.245.88"
          }
        ]
      }
    }
  ]
}
```

Route 53 - Geolocation Routing Policies

Geolocation Routing Policies allow you to direct traffic based on the geographic location of where the request originated from.

For example, this would let you route all traffic coming from North America to servers located in North American regions, where queries from other regions could be directed to servers hosted in that region. (potentially with pricing and language specific to that region)



```
{
  "Changes": [
    {
      "Action": "UPSERT",
      "ResourceRecordSet": {
        "Name": "example.com",
        "Type": "A",
        "SetIdentifier": "US-East",
        "GeoLocationRoutingPolicy": {
          "CountryCode": "US"
        },
        "TTL": 300,
        "ResourceRecords": [
          {
            "Value": "34.229.79.211"
          }
        ]
      }
    },
    {
      "Action": "UPSERT",
      "ResourceRecordSet": {
        "Name": "example.com",
        "Type": "A",
        "SetIdentifier": "EU-West",
        "GeoLocationRoutingPolicy": {
          "CountryCode": "EU"
        },
        "TTL": 300,
        "ResourceRecords": [
          {
            "Value": "3.208.76.48"
          }
        ]
      }
    }
  ]
}
```

Route 53 - Geoproximity Routing Policies

Geoproximity Routing Policies allow you to direct traffic based on the geographic location of your users and your AWS resources.

You can route more or less traffic to a specific resource by specifying a 'Bias' value.

Bias values expand or shrink the size of the geographic region from which traffic is routed to.

You must use Route 53 Traffic Flow to use geoproximity routing policies

The screenshot shows the configuration for a Geoproximity rule. The interface includes a title bar 'Geoproximity rule' with a close button. Below it is a 'Show geoproximity map' button with an information icon. The main configuration area has four sections: 'Region' with a dropdown set to '1', 'Endpoint Location' with a dropdown set to 'US East (N. Virginia)', 'Coordinates' with the text 'Using US East (N. Virginia) Coordinates', and 'Bias' with a text input set to '-2' and a slider ranging from -99 to 99. Each section has an information icon. At the bottom, there is a 'Health checks' section with a right-pointing arrow.

Geoproximity rule

Show geoproximity map

Region: 1

Endpoint Location: US East (N. Virginia)

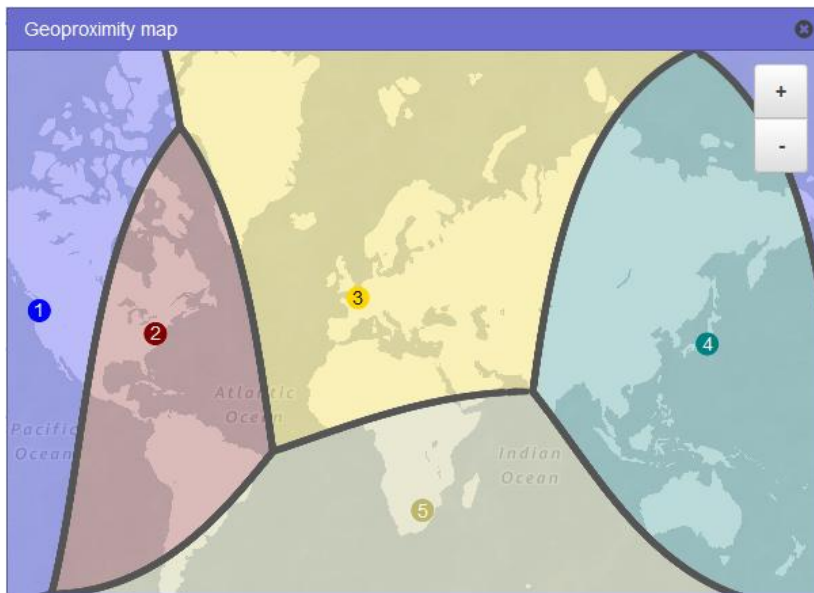
Coordinates: Using US East (N. Virginia) Coordinates

Bias: -2

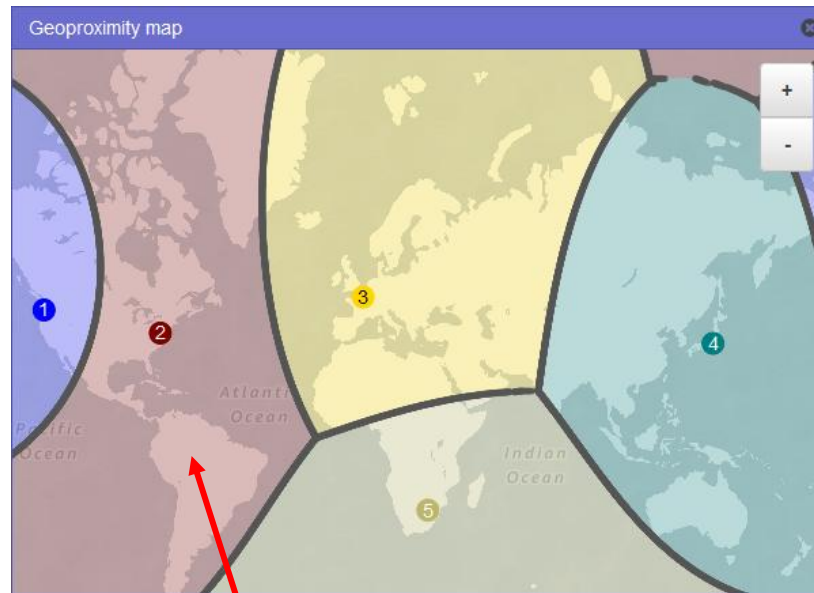
Health checks

Route 53 - Geoproximity Routing Policies

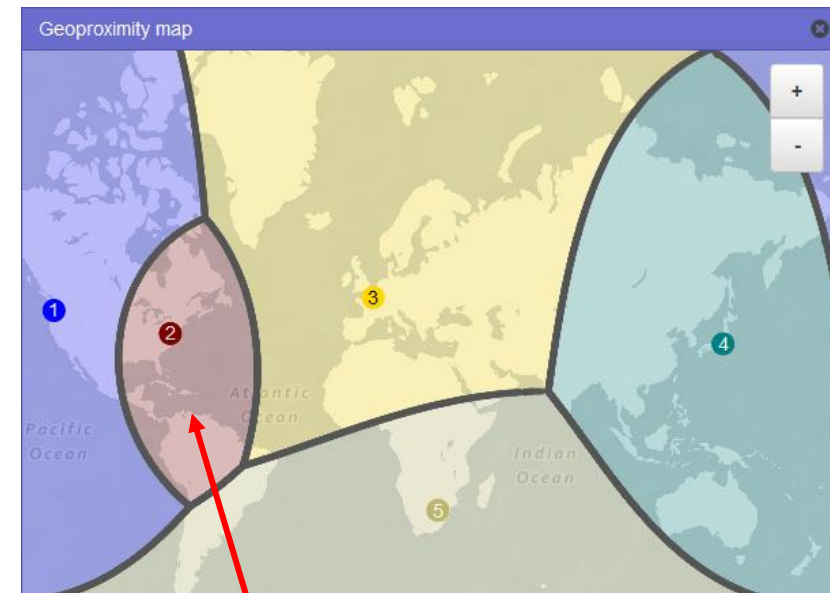
How Bias Works for Geoproximity



Standard Bias



+25 for the US East



-25 for the US East

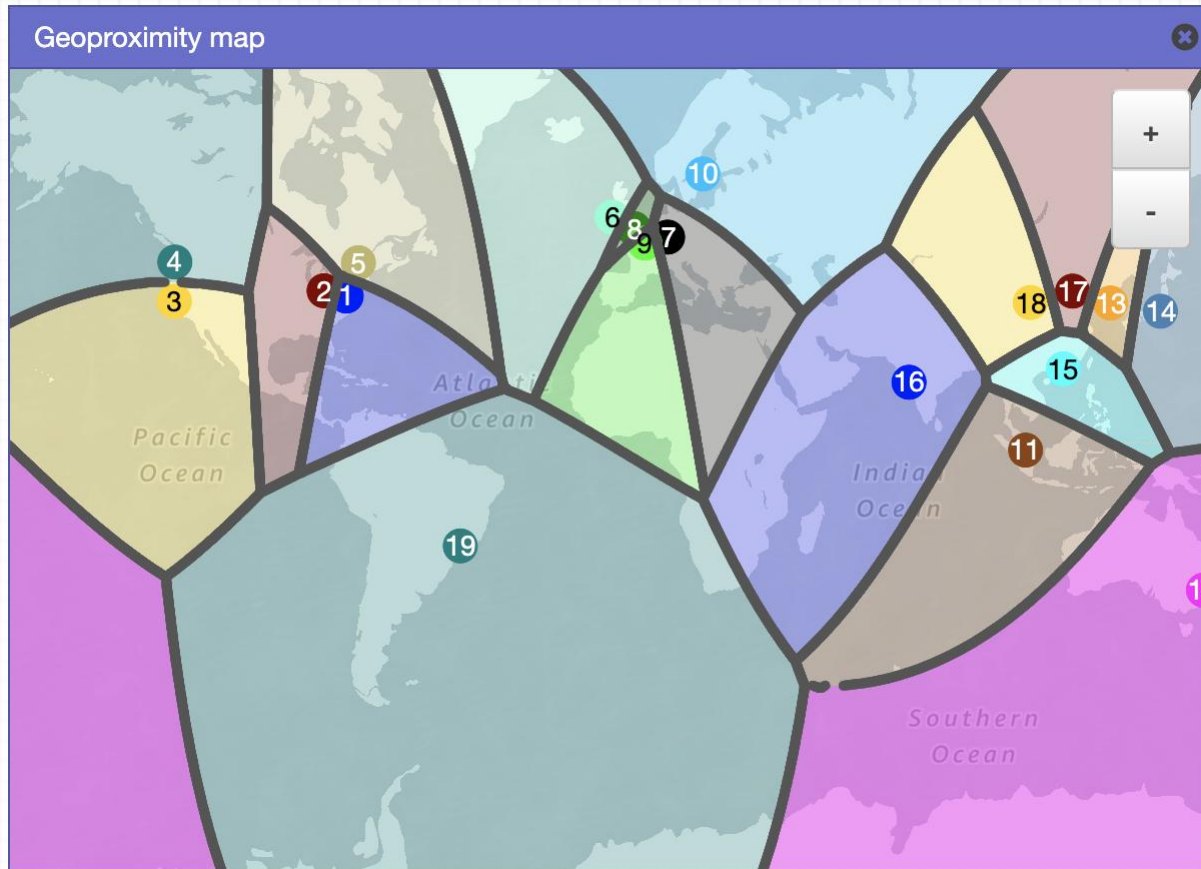
Route 53 – Geoproximity Routing Policies

In the Route 53 Traffic Flow, you can select any region and visualize the bias

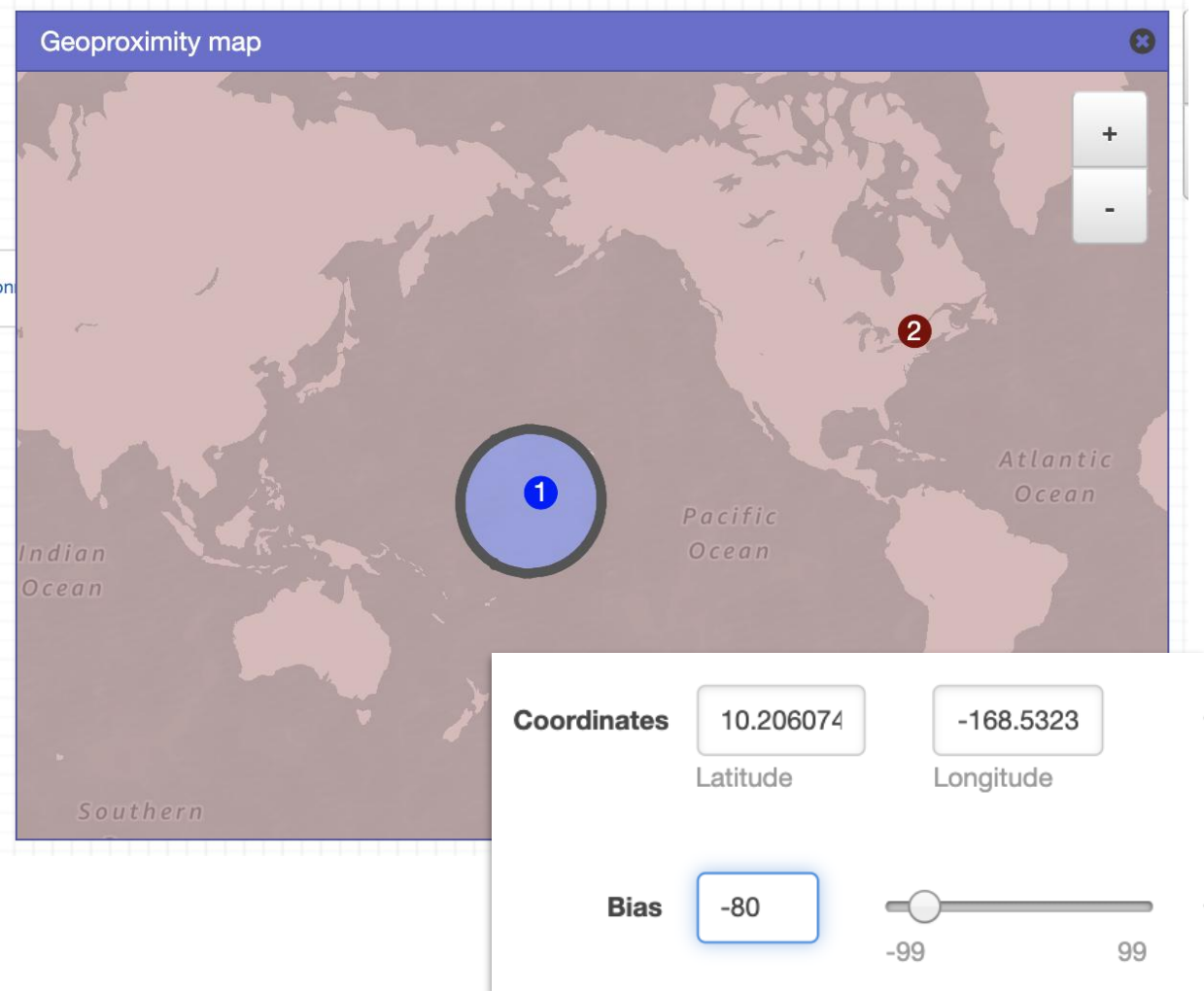
The screenshot displays the Route 53 Traffic Flow console interface. On the left, the 'Geoproximity rule' configuration panel is visible. It includes a 'Hide geoproximity map' button, a 'Region' dropdown set to '1', and an 'Endpoint' dropdown set to 'China (Beijing)'. A 'Location' dropdown is also present, showing a list of regions including EU (Frankfurt), EU (London), EU (Paris), EU (Stockholm), Asia Pacific (Singapore), Asia Pacific (Sydney), Asia Pacific (Seoul), Asia Pacific (Tokyo), Asia Pacific (Hong Kong), Asia Pacific (Mumbai), China (Beijing), China (Ningxia), and South America (São Paulo). The 'Bias' section is also visible. On the right, the 'Geoproximity map' shows a world map with two regions highlighted: a red region labeled '2' in Europe and a blue region labeled '1' in East Asia. The map includes labels for the Pacific Ocean, Atlantic Ocean, Indian Ocean, and Southern Ocean. A zoom control is visible on the right side of the map.

Route 53 - Geoproximity Routing Policies

All Current Regions Selected



Can provide **custom coordinates** over region

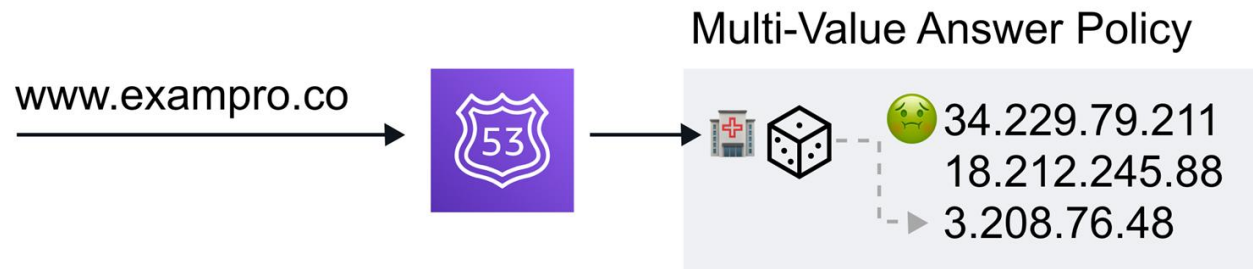


Route 53 - Multi-Value Answer Policies

Multi-Value Answer Policies let you configure Route 53 to return multiple values such as IP addresses for your web servers, in response to DNS queries.

Multiple values can be specified for almost any record. Route 53 automatically performs health checks on resources and only returns values of ones deemed healthy.

Similar to Simple Routing, however, with an added health check for your record set resources.



```
{ "Changes": [{
  "Action": "UPSERT",
  "ResourceRecordSet": {
    "Name": "example.com",
    "Type": "A",
    "TTL": 60,
    "MultiValueAnswerRoutingPolicy": {
      "EvaluateTargetHealth": false
    },
    "ResourceRecords": [
      { "Value": "34.229.79.211" },
      { "Value": "18.212.245.88" },
      { "Value": "3.208.76.48" }
    ]
  }
}] }
```

Route 53 - Health Checks

- Checks health every **30s** by default. Can be reduced to every **10s**
- A health check can **initiate a failover** if the status is returned unhealthy
- A CloudWatch Alarm can be created to alert you of status unhealthy
- A health check can **monitor other health checks** to create a chain of reactions.
- can create up to **50 health checks** for AWS endpoints within or linked to the same AWS account.

	AWS Endpoints	Non-AWS Endpoints
Basic Health Checks	\$0.50* per health check / month	\$0.75 per health check / month
Optional health check features:		
• HTTPS • String Matching • Fast Interval • Latency Measurement	\$1.00 / month per optional feature	\$2.00 / month per optional feature

[Create health check](#) [Delete health check](#) [Edit health check](#) [Refresh](#) [Help](#)

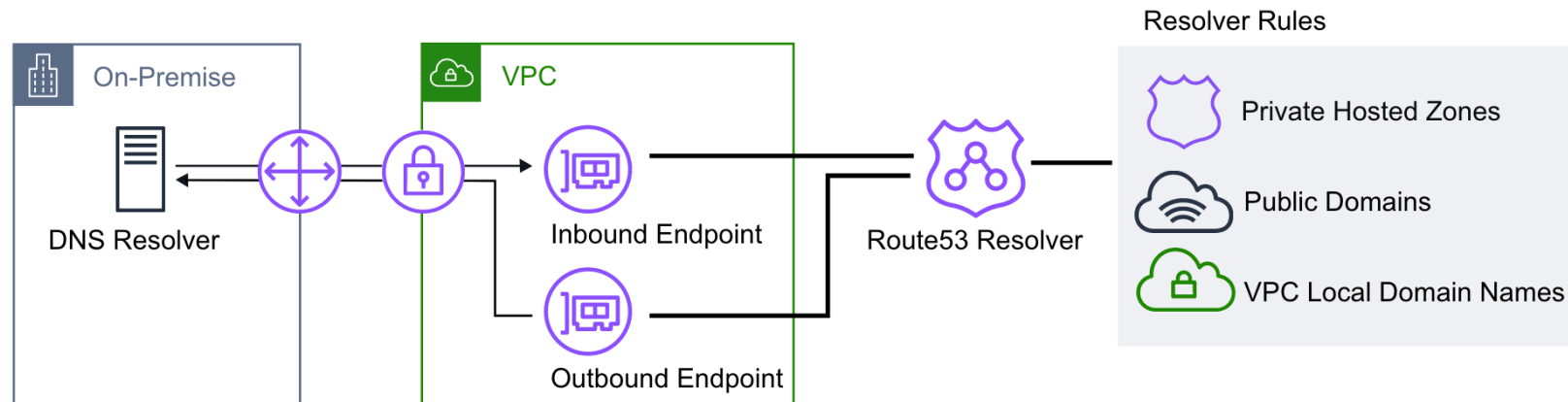
<< < 1 to 1 of 1 health check > >>

	Name	Status	Description	Alarms
☐	app.exampro.co	15 minutes ago now Healthy	https://app.exampro.co:443/health-check	✓ 1 of 1 in OK

Route 53 Resolver

Amazon Route 53 Resolver is a DNS server that allows you to resolve DNS queries between your on-premise network and your VPC

Route53 Resolver was originally known as **.2 resolver and Amazon DNS Server**



- **Inbound Resolver endpoints** allow DNS queries to your VPC from your on-premises network or another VPC.
- **Outbound Resolver endpoints** allow DNS queries from your VPC to your on-premises network or another VPC.

DNSSEC with Route53

Domain Name System Security Extensions (DNSSEC) are a **suite of extension specifications** by the Internet Engineering Task Force (IETF) for **securing data exchanged in the Domain Name System (DNS) in Internet Protocol (IP) networks.**



DNSSEC signing lets DNS resolvers validate that a DNS response came from Amazon Route 53 and has not been tampered with

```
aws route53 create-key-signing-key \
--region us-east-1 \
--hosted-zone-id $hostedzone_id \
--key-management-service-arn $cmk_arn \
--name $ksk_name \
--status ACTIVE \
--caller-reference $unique_string
```

You need to create a KSK signing key and enable it.

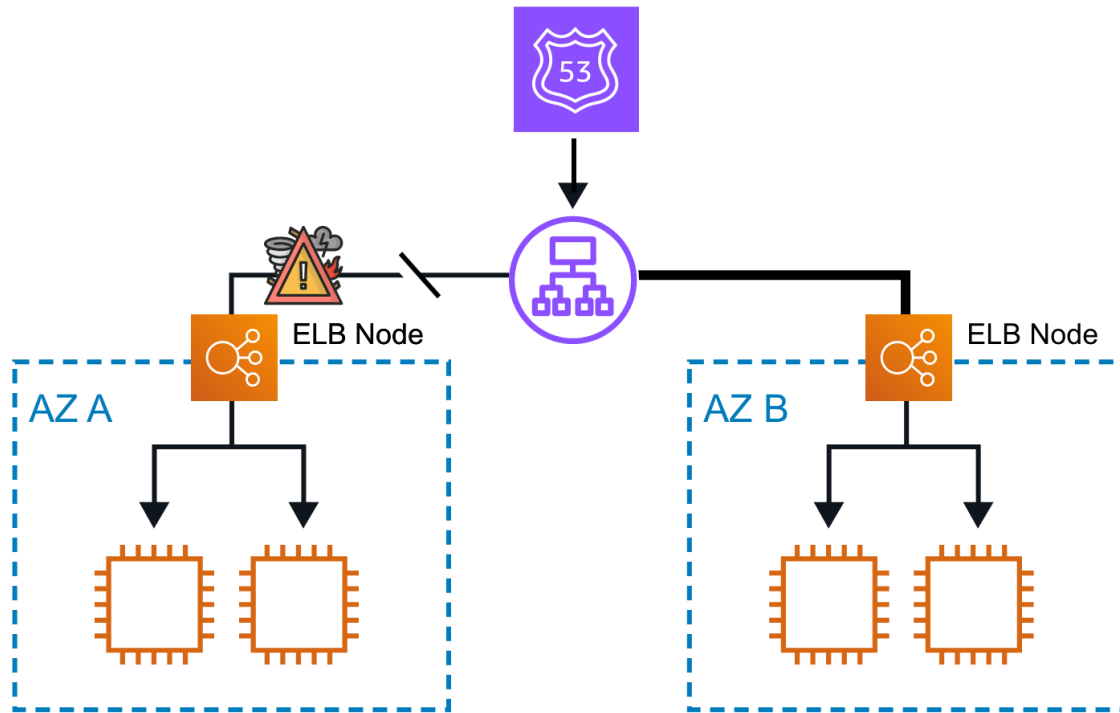
```
aws route53 enable-hosted-zone-dnssec \
--hosted-zone-id $hostedzone_id \
--region us-east-1
```

Enabling DNSSEC is a more involved process than just the two API calls. Its important to enable DNSSEC on domains so people cannot impersonate your domain.

Zonal Shift

Zonal shift is a capability in Amazon Route 53 Application Recovery Controller (Route 53 ARC)

Shifts a load balancer resource away from **an impaired Availability Zone** to healthy AZ with a single action.



- Zonal shifts are only supported on Application Load Balancers and Network Load Balancers with cross-zone load balancing turned off.
- Zonal shift isn't supported when you use an Application Load Balancer as an accelerator endpoint in AWS Global Accelerator.
- You can start a zonal shift for a specific load balancer only for a single Availability Zone. You can't start a zonal shift for multiple Availability Zones

Route 53 Profiles

Route 53 Profiles, lets you apply and manage DNS-related Route 53 configurations across many VPCs and in different AWS accounts.

Resources you can associate with a profile:

- Private hosted zones
- Route 53 Resolver rules
- DNS Firewall rule groups

```
aws route53profiles create-profile --name myprofile
```

```
aws route53profiles associate-resource-to-profile \  
--name test-resource-association \  
--profile-id rp-4987774726example \  
--resource-arn arn:aws:route53resolver:us-east-1:123456789012:firewall-rule-group/rslvr-frg-cfe7f72example \  
--resource-properties "{\"priority\": 102}"
```

```
aws route53profiles associate-profile \  
--name test-association \  
--profile-id rp-4987774726example \  
--resource-id vpc-0af3b96b3example
```